

Panel Data Econometrics Homework: Partial Replication

Replication of Huntington and Liddle (2022), Energy Economics

Master 2 Finance, Technology and Data - Paris 1 Pantheon-Sorbonne

June 2026

Group members: Ethan Benchetrit, Matt Guilbert-Sanderson, Lorenzo Marasco

0. Submission checklist

Required file	Status
Original paper in PDF	Included: Papier.pdf, Huntington and Liddle (2022).
Database used	Included: growth_EE.xlsx, authors replication dataset.
Database formatted for software	Python pipeline constructs the software-ready panel; CSV exports are provided in outputs.
Preferred software code	Python scripts provided: run_replication.py, 01_data_prep.py, 02_estimators.py, 03_exports.py, 04_extension.py.
R and Stata code	R and Stata translated versions are submitted separately in the repository, with comments on implementation difficulties.
Word/TEX and PDF answers	This document is the Word version; the matching PDF version is exported separately.
LLM prompts and answers	Submitted separately as a Word/PDF document containing the LLM prompts and answers used for the homework.
To-do list	Included in Section 17.

Important: the course template states that a commented HTML/code output is not an acceptable substitute for the written answers. This report therefore reproduces the results in a readable narrative and uses the code only as reproducibility support.

1. General identification of the replicated paper

Template item	Answer
APA reference	Huntington, H. G., & Liddle, B. (2022). How energy prices shape OECD economic growth: Panel evidence from multiple decades. <i>Energy Economics</i> , 111, 106082.
Link / DOI	https://doi.org/10.1016/j.eneco.2022.106082
Paper source used here	MPRA/author version uploaded as Papier.pdf; the paper is forthcoming/published in <i>Energy Economics</i> .
Dependent variable Y	Real GDP growth: Delta log real GDP per capita, coded as dlrldpmd or D.GDP.
Preferred explanatory variable X	Energy-price growth: Delta log real energy price index, coded as dlenpr or D.ENERGY.
Panel level	Country-year panel: 18 OECD countries, annual observations.
Dataset	Authors replication dataset growth_EE.xlsx, sheets data and tables5-6.
Key paper correlation	Short-run relationship between energy-price growth and GDP growth, estimated with heterogeneous country slopes and common correlated effects mean-group methods.

2. Abstract and introduction

Abstract

This partial replication studies Huntington and Liddle (2022), who estimate how energy-price movements affect OECD economic growth over several decades. Using the authors panel dataset for 18 OECD countries, we reproduce the descriptive statistics, CIPS unit-root diagnostics, CCEMG estimates, robust mean-group estimates, country-specific responses, and an additional fixed-effects interaction exercise. The replication confirms that energy-price shocks are negatively associated with short-run GDP growth, while the magnitude depends strongly on transformation, specification, cross-sectional dependence, slope heterogeneity, and influential countries. We also add the course-required panel transformations: between, within, first differences, and two-way fixed effects.

Introduction: three findings not fully visible in the paper

First, the replication makes clear that the empirical result depends heavily on the panel transformation used. In levels, energy prices mix cross-country differences and common long-run trends; in first differences and CCEMG specifications, the relevant question becomes whether energy-price shocks are associated with annual GDP-growth movements. This supports the paper focus on changes rather than simple levels.

Second, country heterogeneity is large. The sign and magnitude of the simple energy-growth relationship differ across countries, which justifies heterogeneous-slope estimators such as mean-group and common correlated effects mean-group rather than a pooled model imposing one coefficient for every country.

Third, the dynamic and IV diagnostics must be interpreted cautiously. The template explicitly warns that if the autoregressive parameter is larger than one, long-run multipliers based on $1/(1-\rho)$ are invalid. In this replication, the dynamic IV specification is treated mainly as a diagnostic check; short-run impulse responses are reported, but long-run effects are not over-interpreted when stability is not satisfied.

Rank	Key item found
1	Energy-price shocks are negatively associated with short-run GDP growth in the main CCEMG and robustness specifications, though not in every simple transformation.
2	Cross-country heterogeneity is substantial; the mean effect hides positive, weak, and negative country-specific relationships.
3	The balanced benchmark sample covers 18 OECD countries over 1972-2016, giving 810 country-year observations before differencing/lags.
4	First differences are stationary according to the reported CIPS diagnostics, whereas levels are more persistent.
5	The energy-intensity extension suggests that the marginal effect varies with energy intensity, but the interaction term is not strongly significant.

3. Database update

The original replication dataset is used as the baseline. It contains annual country-level data up to 2016 and is designed to match the definitions and transformations used by Huntington and Liddle. A full update is not performed because the real energy-price series and harmonised macroeconomic variables are constructed from several sources and the precise transformation choices are paper-specific. Updating only GDP or CPI without reconstructing the real energy-price index would create a non-comparable extension. A future update should rebuild the full dataset consistently from Maddison/PWT, OECD or national accounts, and energy-price sources, then rerun every transformation and estimator.

Maximal time span	Frequency	Number of individuals	Max. time obs.	Balanced or unbalanced panel
Original full historical dataset	Annual	18	57 (1960-2016)	Unbalanced: Australia and Germany have missing early years
Benchmark estimation sample	Annual	18	45 (1972-2016)	Balanced: 18 countries x 45 years = 810 observations
Update feasibility check	Not performed as a quantitative extension	Not applicable	Not applicable	A consistent update would require reconstructing the harmonised energy-price index and all macro controls beyond 2016; updating GDP alone would not be comparable.

4. Panel-data sample selection

The dependent variable used for panel-data selection is real GDP growth, Delta log real GDP per capita. The template asks to keep individuals with at least three consecutive observations of the dependent variable. All 18 countries satisfy this condition. The stricter balanced benchmark begins in 1972 because Australia has no usable observations before 1972 and Germany begins later than the other countries. The benchmark sample is therefore a balanced country-year panel from 1972 to 2016.

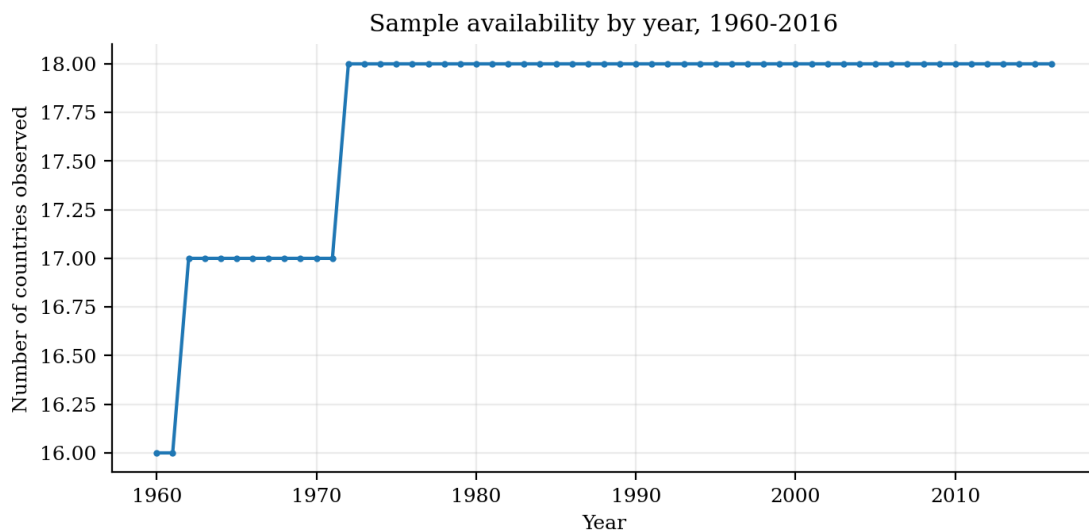
Item	Answer
Individuals excluded	0 countries excluded under the three-consecutive-observation rule. List: none.
Individuals kept	18 countries kept: Australia, Belgium, Canada, Switzerland, Germany, Denmark, Spain, Finland, France, United Kingdom, Ireland, Italy, Japan, Netherlands, Norway, Portugal, Sweden, United States.
Selection-bias comment	The main selection issue is not exclusion but scope: the sample is limited to advanced OECD economies. Poorer and non-OECD countries are excluded by design, not because of missing Y observations.

5. Sample selection within an unbalanced panel

The full historical panel is unbalanced only at the beginning. There is no end-of-sample attrition: all countries are observed up to 2016. Missingness is concentrated in Australia before 1972 and Germany before 1962.

Period	Number of countries observed	Comment
1960-1961	16	Australia and Germany missing
1962-1971	17	Australia missing
1972-2016	18	Balanced benchmark panel

Figure 1. Number of countries observed by year.



Temporal observations	Number of countries	Countries
45	18	Australia, Belgium, Canada, Switzerland, Germany, Denmark, Spain, Finland, France, United Kingdom, Ireland, Italy, Japan, Netherlands, Norway, Portugal, Sweden, United States

Number and proportion of individuals with holes/discontinuities: 0 countries with internal holes inside their observed spell. Missing observations are left-censored starting dates rather than internal breaks; the unbalancedness is therefore delayed data availability, not attrition.

6. Panel-data variables classification

For each variable, the code computes overall, between, and within components. The share of within variance is used to classify variables. All core baseline variables vary with both country and time; no raw baseline variable is strictly individual-invariant. Energy intensity used in the extension is country-specific and effectively time-invariant, so it is wiped out by entity fixed effects unless interacted with a time-varying regressor.

Category	Variables	Count
Varying with two indices	lcpi, lenpr, lrgdpmad, iy, open, expgdp, dlrgdpmad, dlcp, dlenpr, dopen, dexpgdp, diy	K = 12

Category	Variables	Count
Time-invariant	Energy intensity Z_{it} in the extension only; not a baseline regressor in the main CCEMG panel.	$K1 = 1$ if extension included
Individual-invariant	No strictly individual-invariant raw variable; time dummies and cross-sectional averages are generated controls.	$K2 = 0$

Table 4/5. Variable classification sorted by within-variance share.

Variable classification by within-variance share

Variable	N	NT	NT/N	Overall variance	Between variance	Within variance	% within
dopen	18	809	44.9	0.0077	0.0000	0.0077	99.7450
dlenpr	18	809	44.9	0.0054	0.0000	0.0054	99.6215
dexpgdp	18	809	44.9	0.0004	0.0000	0.0004	99.0001
diy	18	809	44.9	0.0002	0.0000	0.0002	98.9493
lcpi	18	810	45.0	0.3656	0.0065	0.3591	98.2237
dlrgdpmad	18	809	44.9	0.0006	0.0000	0.0005	96.1071
dlcpi	18	809	44.9	0.0021	0.0003	0.0018	87.5986
lenpr	18	810	45.0	0.0512	0.0095	0.0417	81.5158
lrgdpmad	18	810	45.0	0.0977	0.0298	0.0679	69.5043
iy	18	810	45.0	0.0015	0.0005	0.0010	66.3393
open	18	810	45.0	0.0934	0.0718	0.0215	23.0541
expgdp	18	810	45.0	0.0082	0.0067	0.0014	17.4557

Notes: baseline sample 1972-2016. Variables sorted by the share of within variance in total variance.

The first-difference variables have very large within-variance shares because country means are small relative to short-run fluctuations. The level variables, especially real GDP and investment, have more persistent between components. This matters because fixed effects identify coefficients from within-country movements, whereas between regressions identify coefficients from cross-country averages.

7. Between and one-way-within descriptive statistics

The next figure plots the distributions of the dependent variable and the preferred explanatory variable after between, within, first-difference and two-way-fixed-effects transformations. Histograms are compared with a normal density with the same two first moments and with a kernel density estimate. The distributions are not perfectly Gaussian and show high-leverage annual shocks, especially for energy-price growth.

Figure 2. Distributions of Y and X by panel transformation.

Distributions by panel transformation: histogram, normal fit and KDE

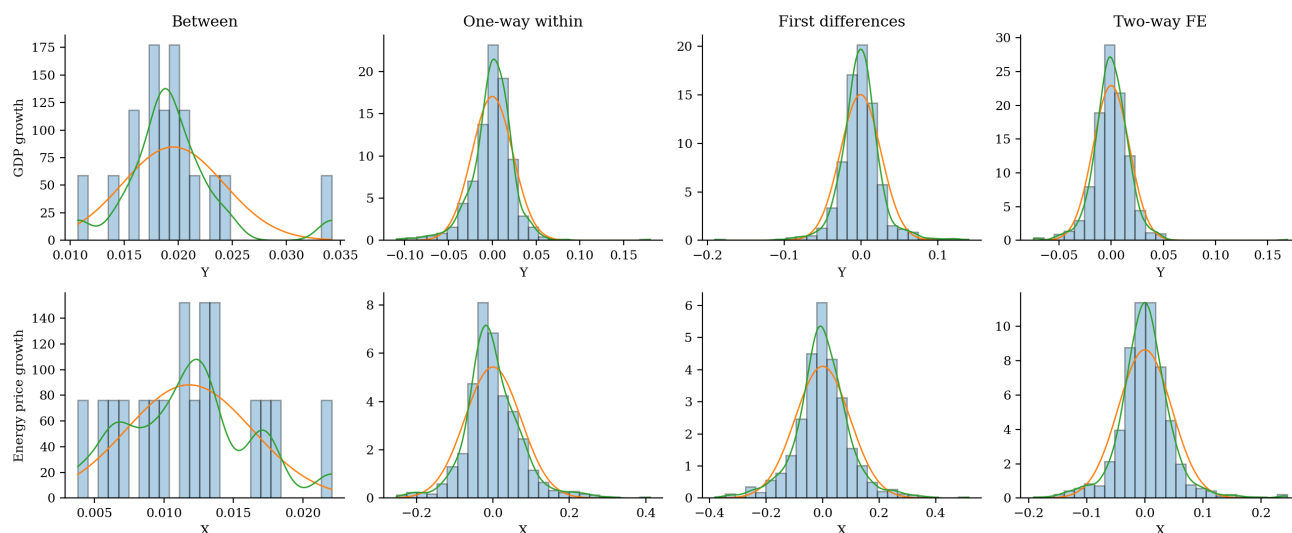


Table. Univariate descriptive statistics for transformed Y and X.

Descriptive statistics by transformation for Y and X

Variable	Transformation	N	Q1	Median	Q3	Mean	Std.Err.	Std. Min	Std. Max	Skew	Kurtosis
Y = D.GDP	Between	810	0.018	0.019	0.021	0.020	0.005	-1.873	3.110	1.278	3.109
Y = D.GDP	One-way within	809	-0.010	0.001	0.013	-0.000	0.023	-4.677	7.688	-0.198	6.218
Y = D.GDP	First differences	791	-0.014	-0.001	0.011	-0.001	0.027	-7.154	5.306	0.094	6.619
Y = D.GDP	Two-way FE	809	-0.009	0.000	0.010	-0.000	0.017	-4.284	9.698	0.798	11.677
X = D.ENERGY	Between	810	0.008	0.012	0.014	0.012	0.005	-1.770	2.262	0.308	-0.301
X = D.ENERGY	One-way within	809	-0.039	-0.007	0.037	0.000	0.073	-3.437	5.599	0.647	3.020
X = D.ENERGY	First differences	791	-0.047	-0.003	0.051	0.000	0.097	-3.919	5.317	0.176	2.687
X = D.ENERGY	Two-way FE	809	-0.023	-0.000	0.022	0.000	0.046	-4.156	5.305	0.422	4.256

8. First differences and two-way fixed effects

First differences are computed within each country after sorting by country and year. The first observation for each country is set to missing, which prevents the first observation of one country from being differenced against the last observation of the previous country. The balanced TWFE transformation is computed as $x_{it} - x_{i.} - x_{.t} + x_{..}$. For the unbalanced version, an equivalent residualisation approach can be used: regress within-transformed variables on time dummies and collect the residuals.

Table. First-difference check for the first observations of each country.

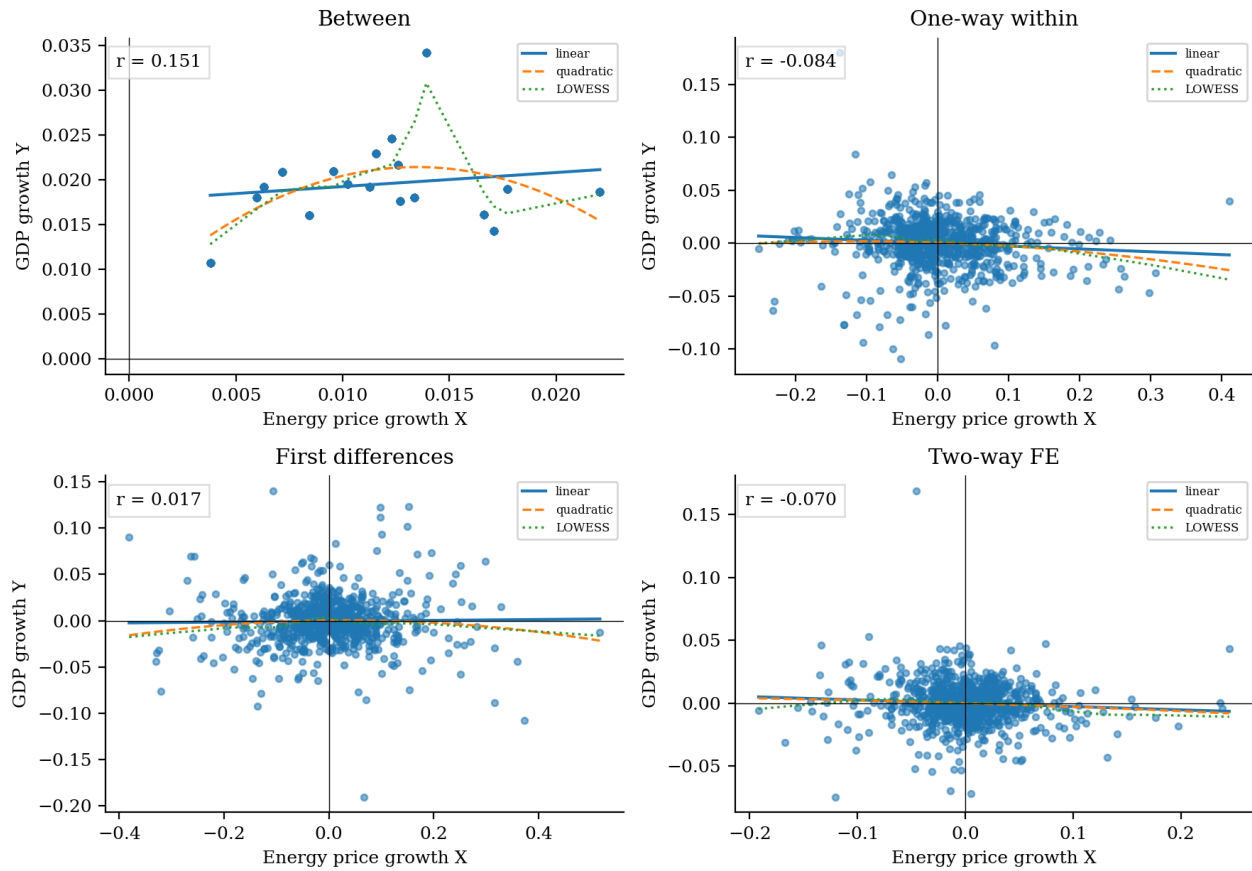
First-difference check: first 30 observations

country	yr	dlogdpmad	dlenpr	FD_Y	FD_X
Australia	1972	.	.	.	.
Australia	1973	0.0375	-0.0501	.	.
Australia	1974	0.0083	-0.0196	-0.0292	0.0304
Belgium	1972	0.0474	-0.1269	.	.
Belgium	1973	0.0564	-0.0158	0.0090	0.1110
Belgium	1974	0.0381	0.0414	-0.0183	0.0572
Canada	1972	0.0398	-0.0756	.	.
Canada	1973	0.0569	0.0522	0.0171	0.1278
Canada	1974	0.0262	0.1459	-0.0307	0.0937
Switzerland	1972	0.0224	-0.0497	.	.
Switzerland	1973	0.0239	-0.0202	0.0015	0.0295
Switzerland	1974	0.0115	0.1442	-0.0123	0.1644
Germany	1972	0.0358	-0.0088	.	.
Germany	1973	0.0414	0.0841	0.0056	0.0929
Germany	1974	0.0080	0.0878	-0.0334	0.0036
Denmark	1972	0.0457	-0.0907	.	.
Denmark	1973	0.0296	-0.0359	-0.0161	0.0547
Denmark	1974	-0.0140	0.3241	-0.0436	0.3601
Spain	1972	0.0873	-0.0531	.	.
Spain	1973	0.0752	-0.0946	-0.0121	-0.0415
Spain	1974	0.0625	0.4227	-0.0126	0.5172
Finland	1972	0.0676	-0.0237	.	.
Finland	1973	0.0592	-0.0043	-0.0084	0.0194
Finland	1974	0.0246	0.2240	-0.0346	0.2284
France	1972	0.0347	0.0155	.	.
France	1973	0.0447	-0.0836	0.0100	-0.0991
France	1974	0.0222	0.1866	-0.0225	0.2703
United Kingdom	1972	0.0317	-0.0352	.	.
United Kingdom	1973	0.0627	-0.0474	0.0310	-0.0122
United Kingdom	1974	-0.0139	0.0082	-0.0767	0.0557

The first transformed observation of each country is missing, preventing accidental differencing across countries.

Figure 3. Bivariate graphs with linear, quadratic and LOWESS fits.

Bivariate graphs with linear, quadratic and LOWESS fits



The bivariate graphs show that the energy-growth relationship is sensitive to transformation. Between variation captures cross-country average differences; within and TWFE variation remove country effects and, in the TWFE case, common year shocks. First differences focus on annual changes and remove deterministic country-level trends more aggressively.

9. Comparison of the four transformed variables

Boxplots by country compare between, one-way within, two-way within and first-difference transformations. Countries differ sharply in the variance of both GDP growth and energy-price growth. This supports the use of heterogeneous-slope estimators and country-by-country diagnostics rather than relying only on pooled averages.

Figure 4. Boxplots by country for transformed GDP growth.

Boxplots by country: GDP growth

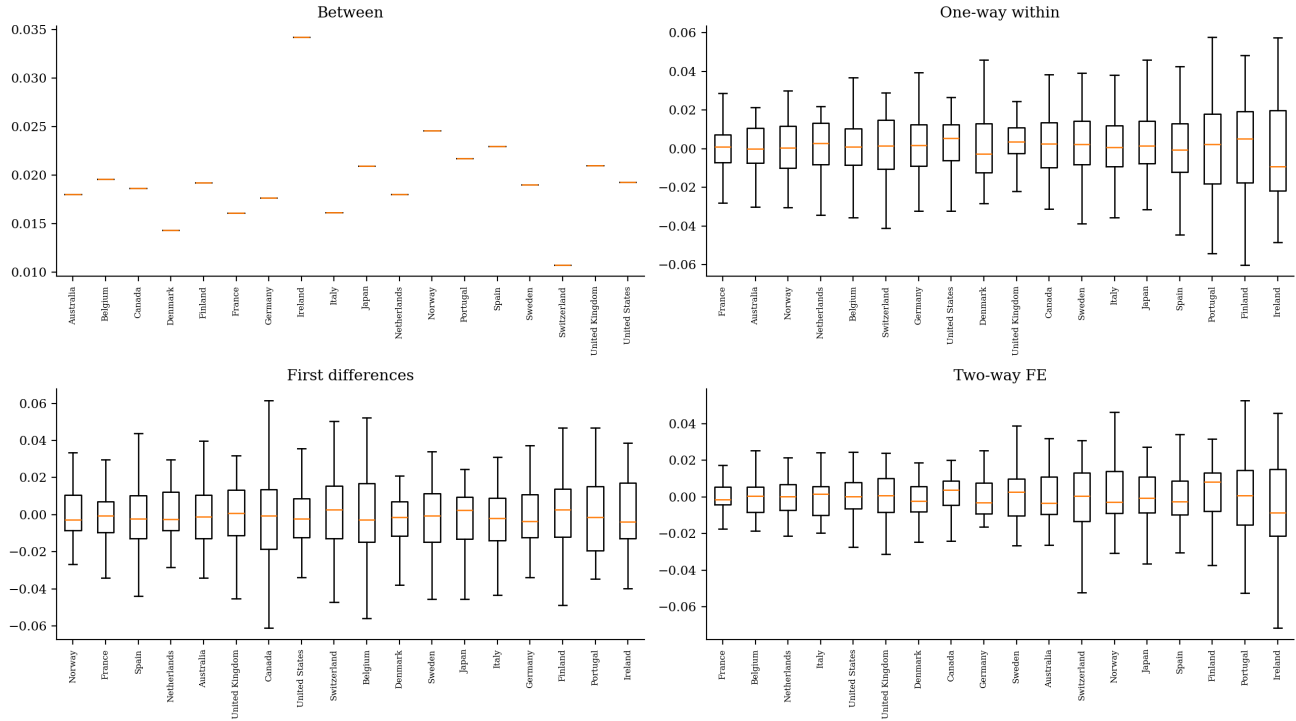
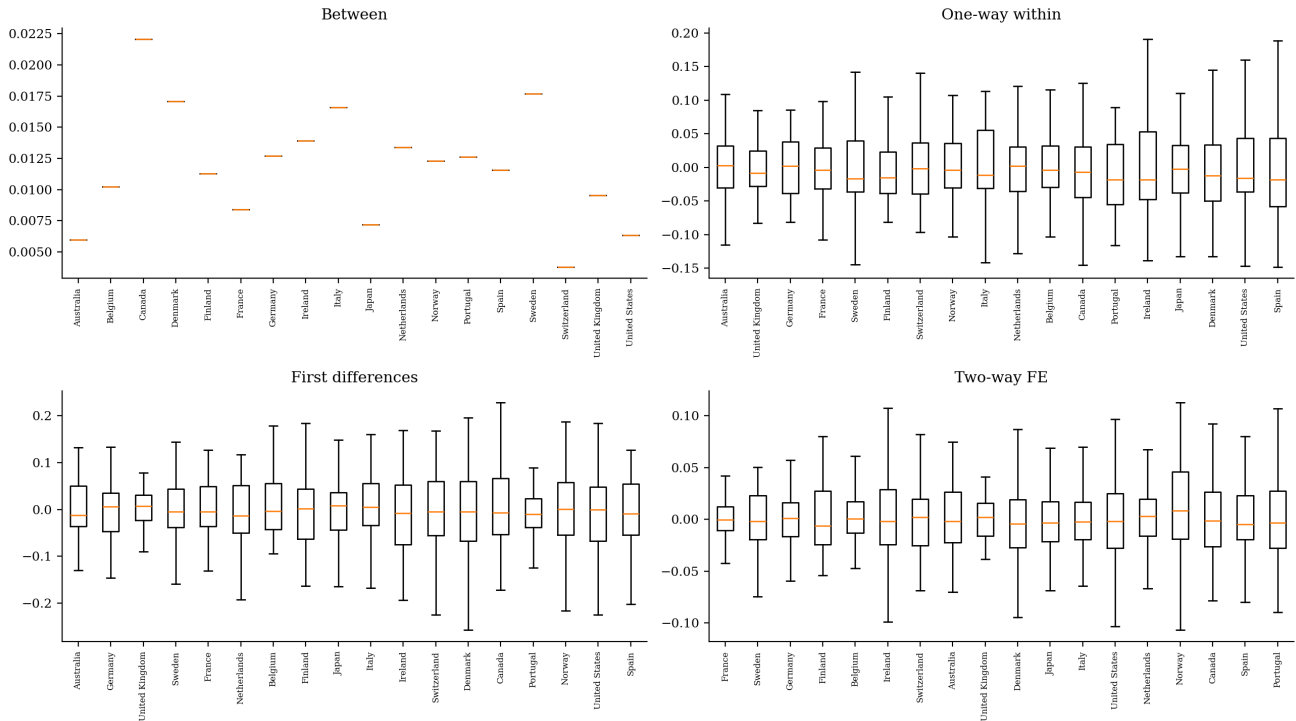


Figure 5. Boxplots by country for transformed energy-price growth.

Boxplots by country: Energy price growth



10. Correlation matrices and autocorrelation diagnostics

The template asks for correlation matrices under between, within, TWFE and first-difference transformations. The following tables report the main six transformed variables. These matrices are used to identify weak simple relationships with the dependent variable and potential multicollinearity among controls. In particular, high correlations among transformed macro controls would justify dropping or re-specifying some variables in finite samples.

Table. Between transformed correlation matrix.

Correlation matrix - between

Variable	D.GDP	D.ENERGY	D.CPI	D.OpenTrade	D.GovExp	D.Invest
D.GDP	1.0	0.151	0.371	0.406	-0.603	0.444
D.ENERGY	0.151	1.0	0.244	0.095	-0.18	0.151
D.CPI	0.371	0.244	1.0	0.11	0.134	-0.123
D.OpenTrade	0.406	0.095	0.11	1.0	-0.578	0.418
D.GovExp	-0.603	-0.18	0.134	-0.578	1.0	-0.722
D.Invest	0.444	0.151	-0.123	0.418	-0.722	1.0

Table. One-way-within transformed correlation matrix.

Correlation matrix - within

Variable	D.GDP	D.ENERGY	D.CPI	D.OpenTrade	D.GovExp	D.Invest
D.GDP	1.0	-0.084	-0.026	0.148	-0.319	0.511
D.ENERGY	-0.084	1.0	0.338	0.165	0.115	-0.041
D.CPI	-0.026	0.338	1.0	0.042	0.193	-0.108
D.OpenTrade	0.148	0.165	0.042	1.0	0.104	-0.067
D.GovExp	-0.319	0.115	0.193	0.104	1.0	-0.368
D.Invest	0.511	-0.041	-0.108	-0.067	-0.368	1.0

Table. Two-way FE transformed correlation matrix.

Correlation matrix - twfe

Variable	D.GDP	D.ENERGY	D.CPI	D.OpenTrade	D.GovExp	D.Invest
D.GDP	1.0	-0.07	-0.109	0.035	-0.2	0.421
D.ENERGY	-0.07	1.0	0.147	-0.007	0.068	-0.086
D.CPI	-0.109	0.147	1.0	-0.011	0.075	-0.07
D.OpenTrade	0.035	-0.007	-0.011	1.0	0.177	-0.162
D.GovExp	-0.2	0.068	0.075	0.177	1.0	-0.285
D.Invest	0.421	-0.086	-0.07	-0.162	-0.285	1.0

Table. First-difference transformed correlation matrix.

Correlation matrix - first diff

Variable	D.GDP	D.ENERGY	D.CPI	D.OpenTrade	D.GovExp	D.Invest
D.GDP	1.0	0.017	-0.155	0.25	-0.226	0.342
D.ENERGY	0.017	1.0	0.453	0.152	-0.003	0.044
D.CPI	-0.155	0.453	1.0	0.121	-0.01	-0.002
D.OpenTrade	0.25	0.152	0.121	1.0	-0.106	0.041
D.GovExp	-0.226	-0.003	-0.01	-0.106	1.0	-0.17
D.Invest	0.342	0.044	-0.002	0.041	-0.17	1.0

In the paper itself, cross-sectional dependence and common shocks are important reasons for using Common Correlated Effects Mean Group estimators. The correlation matrices in this report are therefore descriptive diagnostics rather than a substitute for the CCEMG specification.

11. Investigating bivariate heterogeneity by individuals

The template asks for country-level heterogeneity for first differences and TWFE transformed variables. The tables sort countries by the simple correlation between transformed GDP growth and transformed energy-price growth. The beta column reports the implied simple-regression slope $r(Y,X)*sd(Y)/sd(X)$.

Table. Heterogeneity of the key correlation by country - first differences.

Heterogeneity of key correlation by country - FD

Country	T(t)	r(Y,X)	sd(Y)	sd(X)	sd(Y)/sd(X)	beta	Diagnosis
Switzerland	44	0.411	0.025	0.105	0.238	0.098	positive > 0.08
Germany	44	0.363	0.029	0.067	0.441	0.160	positive > 0.08
Canada	44	0.279	0.024	0.108	0.226	0.063	positive > 0.08
Sweden	44	0.210	0.026	0.074	0.353	0.074	positive > 0.08
Spain	44	0.195	0.020	0.132	0.149	0.029	positive > 0.08
Finland	44	0.132	0.033	0.096	0.348	0.046	positive > 0.08
United States	44	0.123	0.025	0.118	0.210	0.026	positive > 0.08
France	44	0.045	0.018	0.084	0.211	0.010	weak
Australia	43	0.006	0.020	0.058	0.353	0.002	weak
Portugal	44	-0.006	0.035	0.115	0.301	-0.002	weak
Netherlands	44	-0.063	0.020	0.085	0.233	-0.015	weak
Belgium	44	-0.098	0.025	0.094	0.270	-0.026	negative < -0.08
Ireland	44	-0.124	0.046	0.104	0.445	-0.055	negative < -0.08
United Kingdom	44	-0.135	0.023	0.074	0.315	-0.043	negative < -0.08
Norway	44	-0.136	0.016	0.115	0.135	-0.018	negative < -0.08
Japan	44	-0.155	0.027	0.099	0.272	-0.042	negative < -0.08
Denmark	44	-0.226	0.026	0.108	0.241	-0.054	negative < -0.08
Italy	44	-0.364	0.028	0.099	0.279	-0.101	negative < -0.08

Table. Heterogeneity of the key correlation by country - two-way fixed effects.

Heterogeneity of key correlation by country - TWFE

Country	T(i)	r(Y,X)	sd(Y)	sd(X)	sd(Y)/sd(X)	beta	Diagnosis
United Kingdom	45	0.301	0.014	0.045	0.302	0.091	positive > 0.08
Australia	44	0.164	0.016	0.045	0.348	0.057	positive > 0.08
Spain	45	0.134	0.020	0.060	0.324	0.043	positive > 0.08
Germany	45	0.059	0.015	0.033	0.465	0.027	weak
Switzerland	45	0.029	0.016	0.044	0.369	0.011	weak
Canada	45	0.008	0.014	0.059	0.241	0.002	weak
Sweden	45	-0.051	0.016	0.031	0.495	-0.025	weak
Finland	45	-0.065	0.023	0.038	0.598	-0.039	weak
Portugal	45	-0.070	0.023	0.061	0.380	-0.026	weak
Netherlands	45	-0.073	0.010	0.051	0.203	-0.015	weak
Norway	45	-0.163	0.017	0.056	0.296	-0.048	negative < -0.08
United States	45	-0.177	0.013	0.048	0.274	-0.048	negative < -0.08
France	45	-0.189	0.008	0.027	0.306	-0.058	negative < -0.08
Ireland	45	-0.195	0.037	0.043	0.856	-0.167	negative < -0.08
Belgium	45	-0.203	0.010	0.042	0.244	-0.049	negative < -0.08
Japan	45	-0.223	0.018	0.046	0.393	-0.088	negative < -0.08
Italy	45	-0.331	0.012	0.046	0.271	-0.090	negative < -0.08
Denmark	45	-0.398	0.014	0.046	0.303	-0.120	negative < -0.08

Countries are naturally split into positive, negative and weak-correlation groups using the +/-0.08 thresholds suggested in the template. This heterogeneity is a substantive result: it shows that a representative mean effect can coexist with substantial country-specific dispersion.

12. Replication of the paper estimates

This section reports the main replication outputs generated from the Python code. The paper applies mean-group estimators controlling for cross-country correlations. The study reports that averaged across countries, a 10 percent increase in energy prices reduces economic growth by approximately 0.15 percent; our replication focuses on reproducing this negative average relationship and documenting specification sensitivity.

Table 1. Data summary reproduced from the replication code.

Table 1 - Data Summary

Variable	Mean	Standard deviation	Coeff. of variation	Minimum	Maximum
GDP	9.637	0.431	0.045	7.992	10.520
CPI	4.189	0.915	0.219	0.978	5.394
ENERGY	4.335	0.264	0.061	3.467	4.891
OpenTrade	0.518	0.294	0.568	0.089	2.151
GovExp	0.249	0.090	0.360	0.000	0.475
Invest	0.233	0.039	0.166	0.146	0.389
D.GDP	0.024	0.025		-0.090	0.214
D.CPI	0.045	0.042		-0.046	0.321
D.ENERGY	0.006	0.071		-0.242	0.423
D.OpenTrade	0.005	0.082		-1.057	1.016
D.GovExp	0.001	0.019		-0.334	0.120
D.Invest	-0.000	0.014		-0.082	0.117

Notes: Real GDP (GDP), CPI and Energy Prices (ENERGY) are in logarithms. Open Trade (OpenTrade), Gov Expenditures (GovExp) and Investment (Invest) are % of GDP. Overall standard deviations, minimums and maximums are reported. D. denotes change.

Table 2. Pesaran CIPS panel unit-root tests.**Table 2 - Pesaran Panel Unit Root Tests**

Variable	Lag 0	Lag 1	Lag 2	Lag 3
No Trend				
GDP	-1.43	-1.50	-1.28	-1.34
CPI	-1.90	-2.59**	-1.83	-1.65
ENERGY	-2.06*	-1.94	-1.82	-1.68
OpenTrade	-2.20*	-2.29*	-2.05*	-1.86
GovExp	-1.85	-1.74	-1.74	-1.81
Invest	-1.96	-2.24*	-2.14*	-1.95
D.GDP	-5.64**	-4.82**	-4.27**	-3.99**
D.CPI	-4.00**	-4.14**	-3.93**	-3.63**
D.ENERGY	-7.91**	-6.40**	-5.81**	-5.20**
D.OpenTrade	-6.47**	-5.70**	-5.06**	-3.89**
D.GovExp	-7.47**	-5.11**	-4.31**	-3.94**
D.Invest	-6.00**	-5.07**	-4.47**	-4.10**
Trend				
GDP	-1.81	-2.01*	-1.81	-1.84
CPI	-2.71**	-3.76**	-2.84**	-2.56*
ENERGY	-2.71**	-2.61**	-2.41*	-2.26*
OpenTrade	-2.32*	-2.41*	-2.17*	-1.92
GovExp	-2.38*	-2.29*	-2.34*	-2.43*
Invest	-2.10*	-2.38*	-2.27*	-2.16*

Notes: * $p < 0.05$; ** $p < 0.01$. D. = first-difference. Null: all panels have a unit root. The statistic reported is the raw t-bar.

The CIPS table reports raw t-bar statistics, matching the convention used in the replication code. The normal-approximation stars should be read as indicative rather than as a full replacement for Pesaran critical values. The economically important conclusion is that variables in first differences reject the unit-root null much more clearly than most variables in levels.

Table 3. CCE-MG: unweighted and outlier-robust estimates.

Table 3 - CCE-MG: Unweighted vs outlier-robust estimates

Variable	cce	ccerobust
D.CPI	-0.209**	-0.252**
	(0.056)	(0.043)
D.Energy	-0.018	-0.013
	(0.013)	(0.011)
D.OpenTrade	0.108**	0.126**
	(0.041)	(0.033)
D.GovExp	-0.123	-0.108
	(0.069)	(0.060)
D.Invest	0.515**	0.521**
	(0.110)	(0.095)
L.GDP	-0.201**	-0.154**
	(0.039)	(0.028)
L.CPI	-0.083**	-0.091**
	(0.031)	(0.027)
L.OpenTrade	0.094*	0.080**
	(0.039)	(0.031)
L.Invest	0.164	0.169*
	(0.086)	(0.072)
_cons	0.348	0.328*
	(0.180)	(0.160)

Notes: b/(se) format. * $p < 0.05$; ** $p < 0.01$. D. = change; L. = lag.

Table 4. CCE mean-group estimates for real GDP growth.

Table 4 CCE-Mean-group estimates for real GDP growth

Variable	inst	exog	w/o_mod	recession	ger_ire	sixties
L.D.CPI>2%	-0.222**	-0.054	-0.133	-0.121	-0.222**	-0.167*
	(0.062)	(0.081)	(0.070)	(0.124)	(0.062)	(0.068)
D.OpenTrade	0.089	0.116*	0.077*	0.136**	0.089	0.088*
	(0.054)	(0.055)	(0.039)	(0.051)	(0.054)	(0.040)
D.GovExp	-0.044	-0.075	-0.143*	-0.059	-0.044	-0.066
	(0.068)	(0.059)	(0.068)	(0.066)	(0.068)	(0.054)
D.Invest	0.649**	0.593**	0.627**	0.657**	0.649**	0.542**
	(0.156)	(0.143)	(0.144)	(0.162)	(0.156)	(0.112)
L.GDP	-0.397**	-0.435**	-0.373**	-0.417**	-0.397**	-0.217**
	(0.074)	(0.081)	(0.067)	(0.077)	(0.074)	(0.046)
L.CPI	-0.019	-0.035	-0.026	-0.030	-0.019	-0.014
	(0.071)	(0.061)	(0.069)	(0.066)	(0.071)	(0.042)
L.OpenTrade	0.196**	0.225**	0.126**	0.189**	0.196**	0.124**
	(0.054)	(0.065)	(0.041)	(0.058)	(0.054)	(0.036)
L.Invest	0.618**	0.624**	0.415**	0.616**	0.618**	0.222*
	(0.175)	(0.179)	(0.156)	(0.193)	(0.175)	(0.101)
mod	0.004	0.000		0.004	0.004	0.001
	(0.007)	(0.006)		(0.007)	(0.007)	(0.004)
D.Energy×mod	-0.065	-0.066		-0.067	-0.065	-0.046
	(0.082)	(0.039)		(0.075)	(0.082)	(0.051)
D.Energy	0.008		-0.027	-0.024	0.008	-0.028
	(0.059)		(0.022)	(0.057)	(0.059)	(0.035)
recess				0.003		
				(0.007)		
_cons	0.065	0.133	-0.007	-0.036	0.065	-0.055
	(0.412)	(0.428)	(0.430)	(0.626)	(0.412)	(0.169)
Price effect (post-1982)	-0.057		-0.027	-0.092	-0.057	-0.075

Notes: b/(se) format. * p<0.05; ** p<0.01. inst=IV; exog=exogenous; w/o_mod=no split; recession=2009 dummy; ger_ire=excl outliers; sixties=incl 1960.

The CCEMG results show a robust role for macro controls and a negative post-1982 price effect in the robustness table. The coefficient on energy-price growth is sensitive to specification and to the Great Moderation split, consistent with the paper emphasis on historical regimes and heterogeneous country responses.

Table 5. Individual country energy intensity and responses.

Table 5 Individual country energy intensity and responses

Country	Intensity	Exports	Imports	Post-1982	Pre-1983
Australia	6.617	9.462	4.660	-0.041	0.149
Belgium	6.230	0.490	2.477	-0.063	-0.204
Canada	10.861	16.607	12.377	0.042	0.004
Switzerland	3.072	0.605	1.230	0.021	-0.109
Germany	4.224	5.535	14.295	-0.142	-0.096
Denmark	3.614	0.738	0.817	0.012	0.149
Spain	3.996	1.488	5.077	0.032	0.046
Finland	6.438	0.419	1.184	-0.097	-0.231
France	4.697	4.758	10.227	-0.003	0.221
United Kingdom	4.615	8.877	9.302	0.038	-0.102
Ireland	3.274	0.074	0.506	0.067	0.524
Italy	3.390	1.311	7.116	0.006	-0.252
Japan	4.542	3.618	20.419	-0.091	-0.108
Netherlands	5.769	2.729	3.772	-0.009	-0.349
Norway	7.840	8.174	1.790	-0.095	-0.112
Portugal	3.369	0.154	0.903	0.248	-0.102
Sweden	6.607	1.392	2.232	-0.124	0.564
United States	7.221	72.613	91.547	-0.073	0.152
Average				-0.015	0.008

Notes: Post-1982 coef = D.Energy + D.Energy×mod. Pre-1983 coef = D.Energy only. Intensity uses sheet table5-6 when available.

Table 6. Regressions of country-specific responses on energy intensity.**Table 6 Regressions of country-specific responses on energy intensity**

Specification	Coef	t-stat	N	F	RMSE
(1) Pre-1983 ~ Intensity	0.000	0.00	18	0.00	0.242
(2) Post-1982 ~ Intensity	-0.015	-1.06	18	1.12	0.083
(3) Pre-1983 ~ Intensity	0.016	0.91	16	0.83	0.217
(4) Post-1982 ~ Intensity	-0.016	-1.06	16	1.13	0.079

Notes: Intensity = energy use per unit of GDP (EIA). * p<0.05; ** p<0.01. Robust standard errors.

13. Standard panel estimates requested by the template

The course template asks for a single table comparing Between, Within, Mundlak, two-way fixed effects and first differences. These estimators are not the main estimators of Huntington and Liddle, but they are useful diagnostics because they show how the coefficient on energy-price growth changes when identification comes from cross-country means, within-country deviations, common-shock-adjusted deviations, or annual first differences.

Table. Standard panel estimates requested by the course template.

Panel estimates: Between, Within, Mundlak, TWFE and FD

Variable	Between	Within FE	Mundlak pooled RE	Two-way FE	First differences
dlenpr	-0.100 (0.189)	-0.035** (0.013)	-0.035** (0.013)	-0.006 (0.014)	0.017 (0.013)
dlcpi	0.146** (0.044)	0.047* (0.020)	0.047* (0.020)	-0.054* (0.027)	-0.254** (0.064)
dopen	-0.058 (0.188)	0.056* (0.027)	0.056* (0.027)	0.026 (0.019)	0.049 (0.029)
dexpgdp	-1.589** (0.425)	-0.202 (0.116)	-0.202 (0.115)	-0.093 (0.093)	-0.161 (0.113)
dly	0.164 (0.668)	0.774** (0.084)	0.774** (0.087)	0.560** (0.070)	0.465** (0.087)
N	18	809	809	809	791

Notes: b/(robust se). Mundlak includes country-specific means of regressors; TWFE is obtained by residualizing country and year means.

Mundlak is implemented as a pooled random-effects-style specification augmented with country means of the time-varying regressors. This makes explicit whether cross-country average differences in the controls matter in addition to within-country movements.

14. Dynamic panel diagnostics and Anderson-Hsiao-type IV

Because the dependent variable and the preferred explanatory variable are persistent macroeconomic series, the template asks for a dynamic ARDL / Anderson-Hsiao-type diagnostic when first-difference autocorrelation remains non-negligible. The dynamic specification includes lagged GDP growth, current and lagged energy-price growth, and differenced controls. Second lags in levels are used as excluded instruments in the IV diagnostic.

Table. Dynamic OLS and Anderson-Hsiao-type IV estimates.

Dynamic ARDL / Anderson-Hsiao-type estimates

Variable	Dynamic OLS	IV2SLS (AH-type)
lag_dY	0.186** (0.046)	
dlenpr	-0.024 (0.013)	
lag_dX	-0.035** (0.011)	
dlcpi	0.028 (0.019)	
dopen	0.059* (0.027)	
dexpgdp	-0.167 (0.116)	
dly	0.637** (0.082)	
N	774	not estimated

Notes: b/(robust se). IV uses second lags in levels as instruments. Interpretation is diagnostic if the autoregressive parameter is not stable.

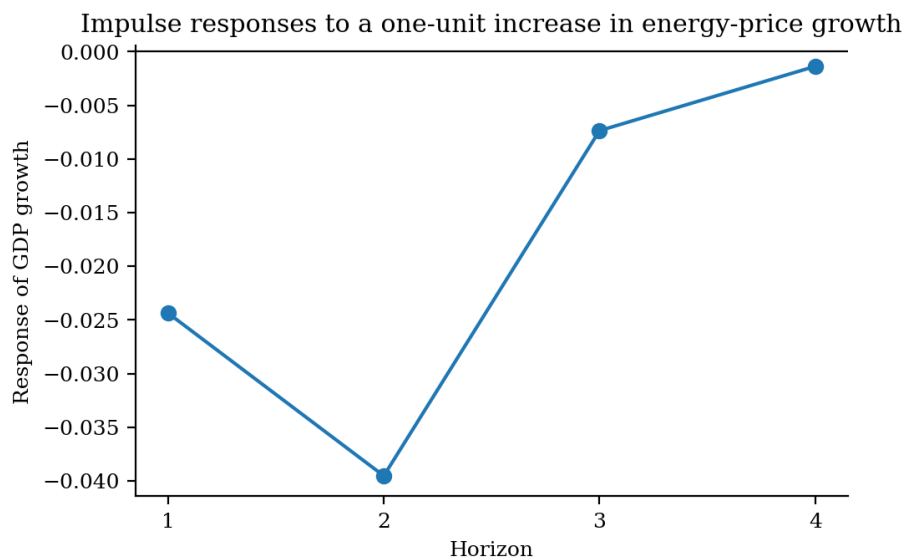
Table. First-stage diagnostics for the dynamic IV exercise.

Dynamic IV first-stage diagnostics

Endogenous/checked variable	N	R2	Partial signal comment
lag_dY	774	0.219	mechanical or sufficient raw R2
dlenpr	774	0.239	mechanical or sufficient raw R2
lag_dX	774	0.136	mechanical or sufficient raw R2

Notes: preliminary OLS first-stage diagnostics using second lags in levels as instruments and current differenced controls.

Figure. Impulse responses to a one-unit increase in energy-price growth.

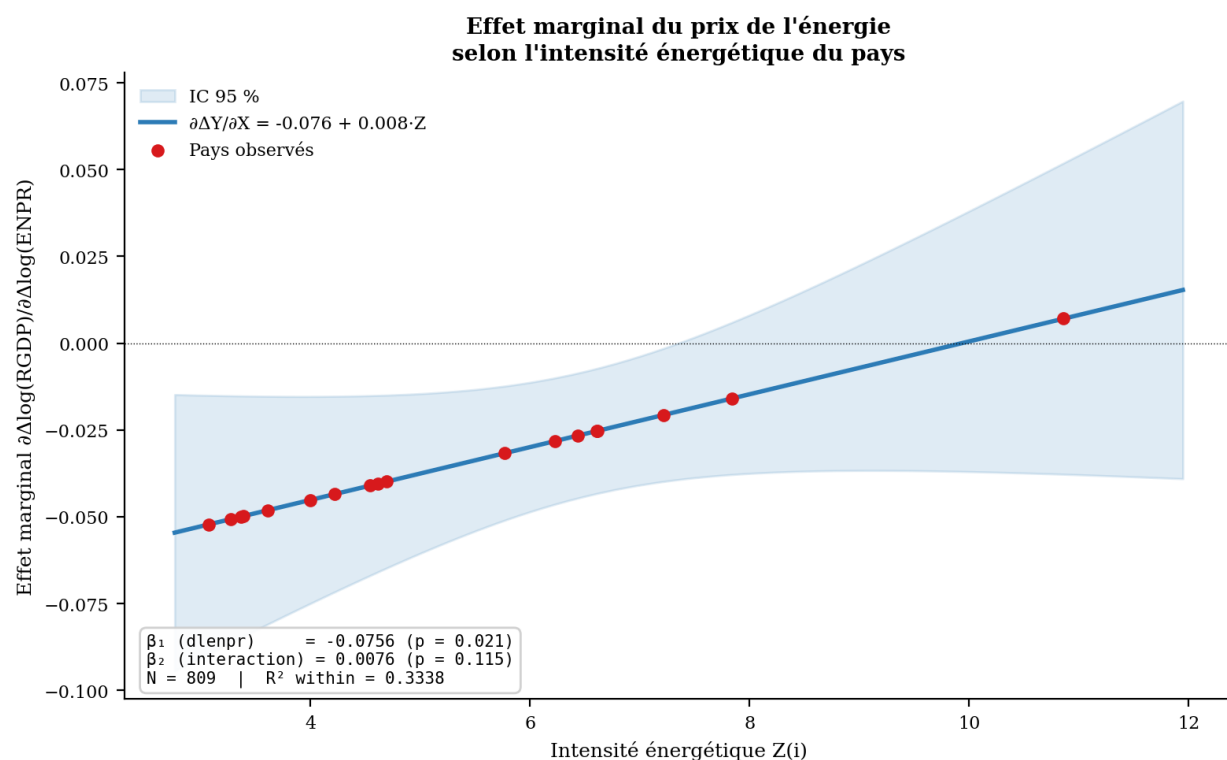


The long-run coefficient is computable because the estimated autoregressive parameter is below one in absolute value. The computed value is -0.073.

15. Own estimation not done in the paper: FE interaction with energy intensity

As an original extension, we estimate a country fixed-effects model interacting the energy-price shock with a time-invariant energy-intensity variable Z_i . The model is $\Delta \log(\text{RGDP})_{it} = \alpha_i + \beta_1 \text{dlenpr}_{it} + \beta_2 (Z_i \text{dlenpr}_{it}) + \text{controls} + \text{error}_{it}$. Since Z_i is time-invariant, its level is absorbed by country fixed effects, but its interaction with the time-varying energy-price shock remains identified.

Figure. Marginal effect of energy-price growth by energy intensity.



The estimated baseline marginal effect is negative at low energy-intensity levels. The interaction slope is positive in this implementation and not statistically significant at conventional levels, meaning that the extension should be interpreted as suggestive rather than definitive. This is still useful because it translates the template advice on time-invariant variables into an estimable interaction term.

Variable	Coef	Std.Err.	t-stat	p-value
Intercept	0.0182	0.0012	14.6194	0.0000
dlenpr	-0.0756	0.0327	-2.3100	0.0211
interaction	0.0076	0.0048	1.5788	0.1148
dlcpi	0.0485	0.0283	1.7149	0.0868
dopen	0.0563	0.0224	2.5137	0.0121
dexpgdp	-0.1980	0.1270	-1.5586	0.1195
diy	0.7719	0.1222	6.3189	0.0000

16. References cited in the text

- Huntington, H. G., & Liddle, B. (2022). How energy prices shape OECD economic growth: Panel evidence from multiple decades. *Energy Economics*, 111, 106082.
- Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4), 967-1012.
- Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265-312.
- Anderson, T. W., & Hsiao, C. (1981). Estimation of dynamic models with error components. *Journal of the American Statistical Association*, 76(375), 598-606.
- Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6), 1417-1426.
- Wansbeek, T., & Kapteyn, A. (1989). Estimation of the error-components model with incomplete panels. *Journal of Econometrics*, 41(3), 341-361.

17. Appendix: ordered to-do list for future panel-data studies

Step	Action
1. Define unit and time index	Identify whether the panel is country, firm, bank, sector, household or asset-level; verify unique (i,t) keys.
2. Define Y and preferred X	Write variable definitions in economic words and in code names before estimating.
3. Audit raw data	Check missing values, duplicates, outliers, units, logs, ratios and source changes.
4. Select panel sample	Apply at least three consecutive observations of Y for dynamic checks; document excluded units.
5. Balanced/unbalanced diagnostics	Report N by date, T by individual, holes and attrition patterns.
6. Classify variables	Compute between and within variances; identify time-invariant and individual-invariant variables.
7. Transform variables	Compute between, within, first differences and TWFE transformations carefully within individual.
8. Plot distributions	Compare histogram, normal fit and KDE for Y and preferred X under each transformation.
9. Examine correlations	Compare between, within, TWFE and FD matrices; inspect autocorrelations and trend correlations.
10. Check heterogeneity	Compute country/individual-specific correlations and slopes before imposing pooled models.
11. Estimate baseline models	Report between, within FE, Mundlak, TWFE and FD models with comparable controls.
12. Handle endogeneity	Justify instruments; report first stages, weak-instrument diagnostics and sensitivity.
13. Dynamic specification	If persistence remains, estimate ARDL or Anderson-Hsiao diagnostics and avoid invalid long-run multipliers.
14. Robustness and extensions	Test regime splits, outlier exclusions, alternative controls and theoretically motivated interactions.
15. Reproducibility	Provide raw data, cleaned data, code, outputs, environment requirements and LLM prompts as separate deliverables.